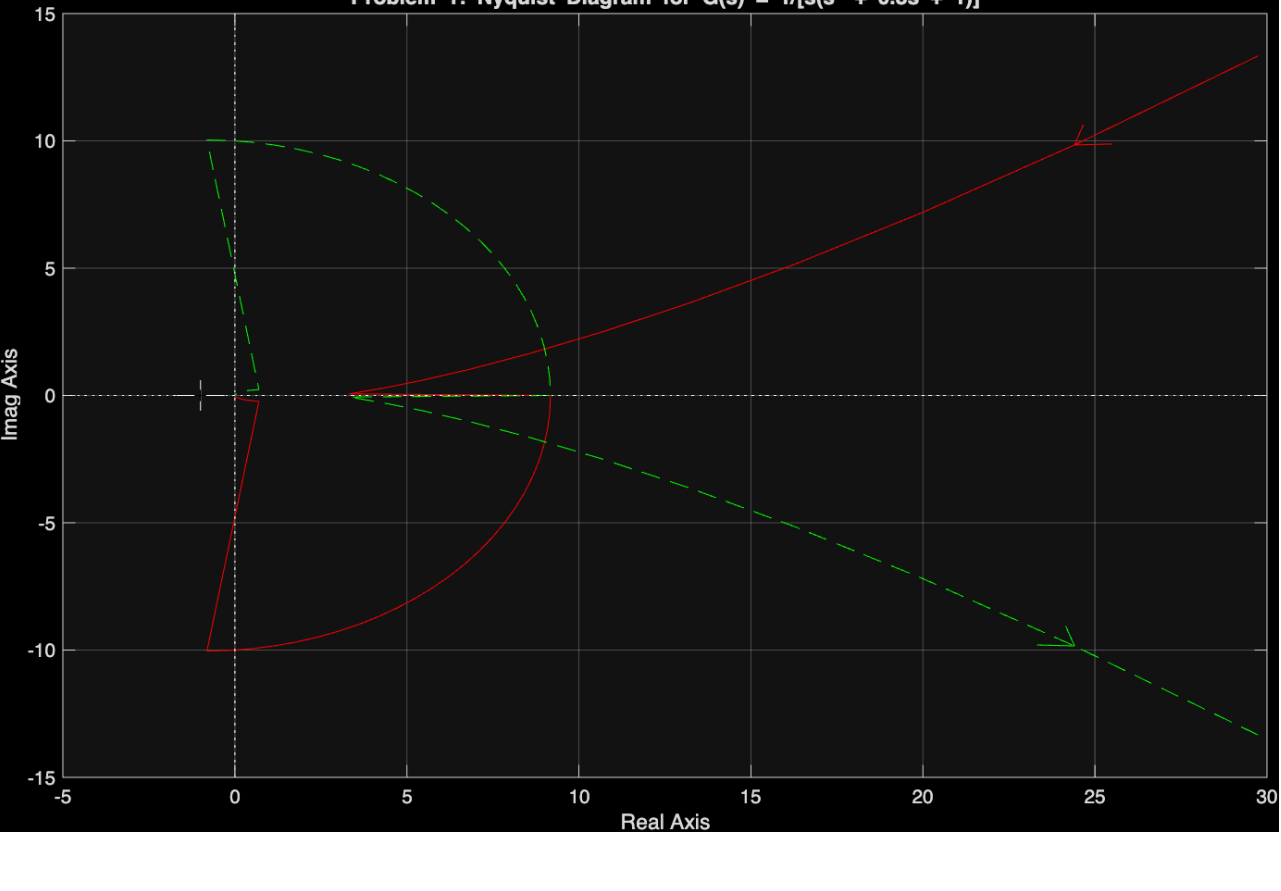


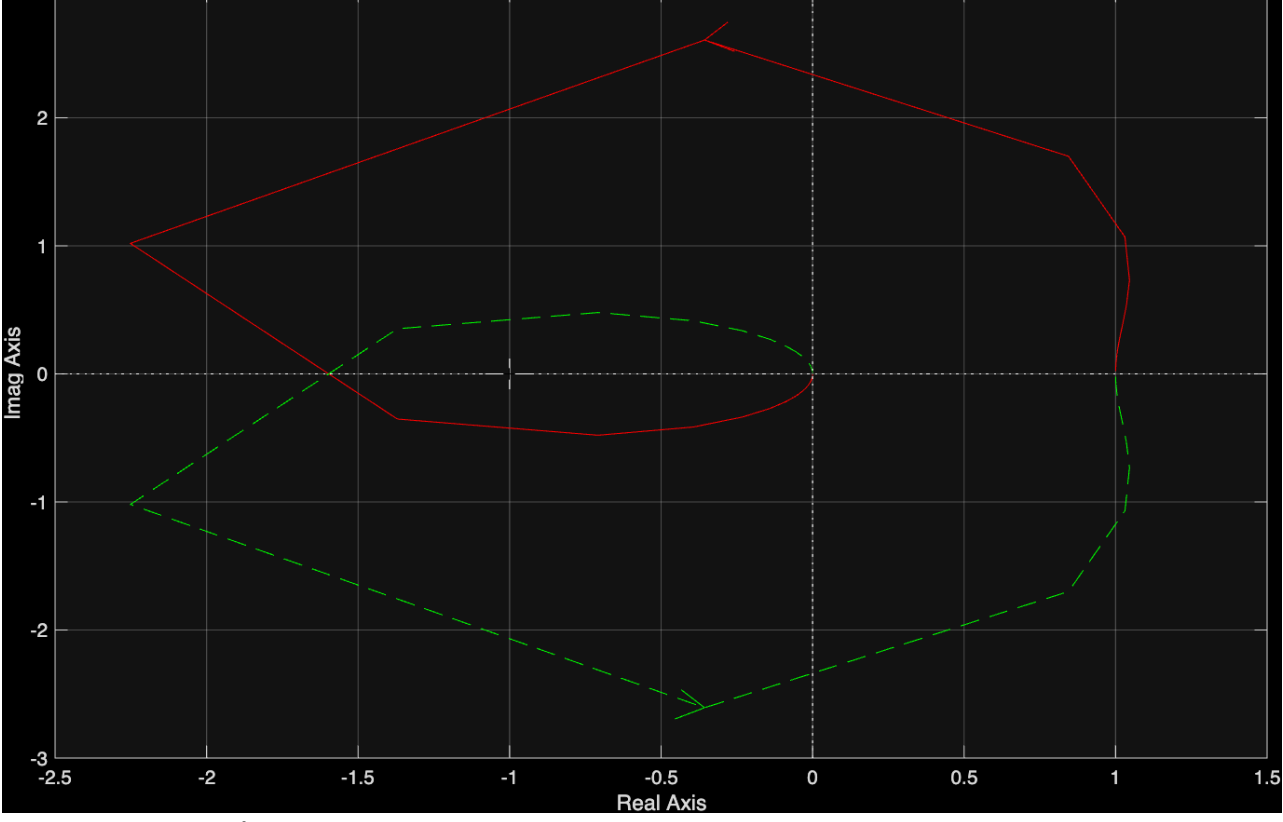
HW6

12/6/25 4:48 PM

1. The system is stable because there is no right half plane poles and the nyquist diagram does not encircle -1



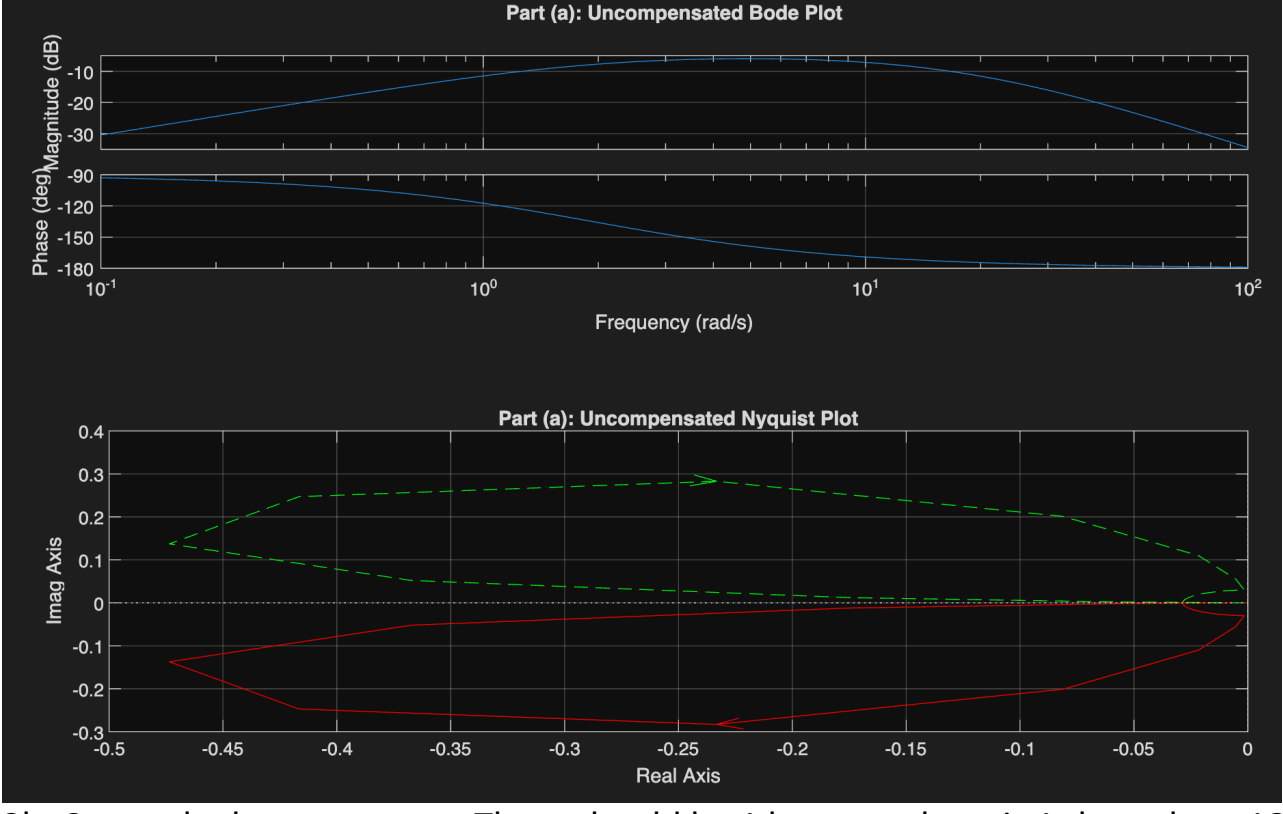
1.



2. Nyquist diagram

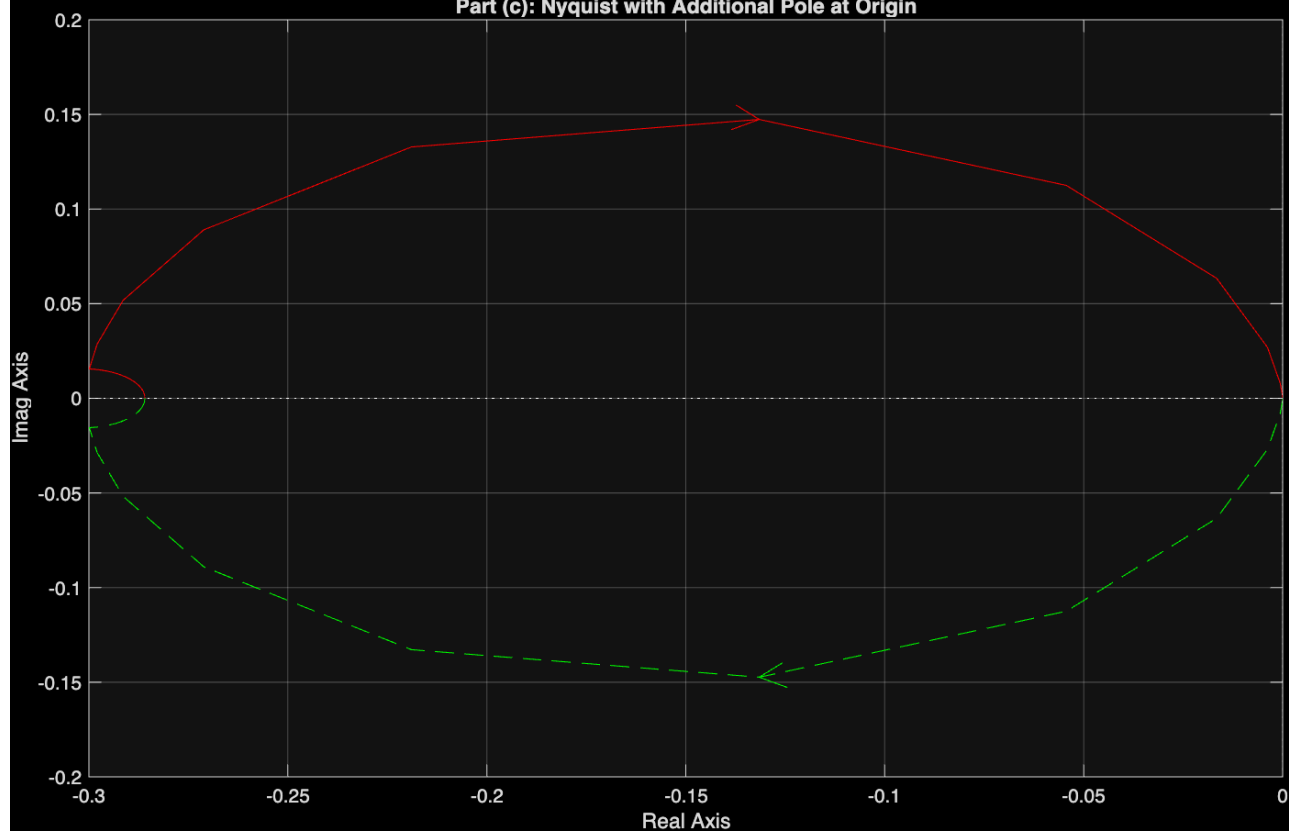
N = -2. cw encircle ments of -1
P = 2 rhp poles: (
0.2623 + 1.1451i
0.2623 - 1.1451i)
Z = 2-2 = 0 which means this system is stable

3a Uncompensated Bode and Nyquist plots



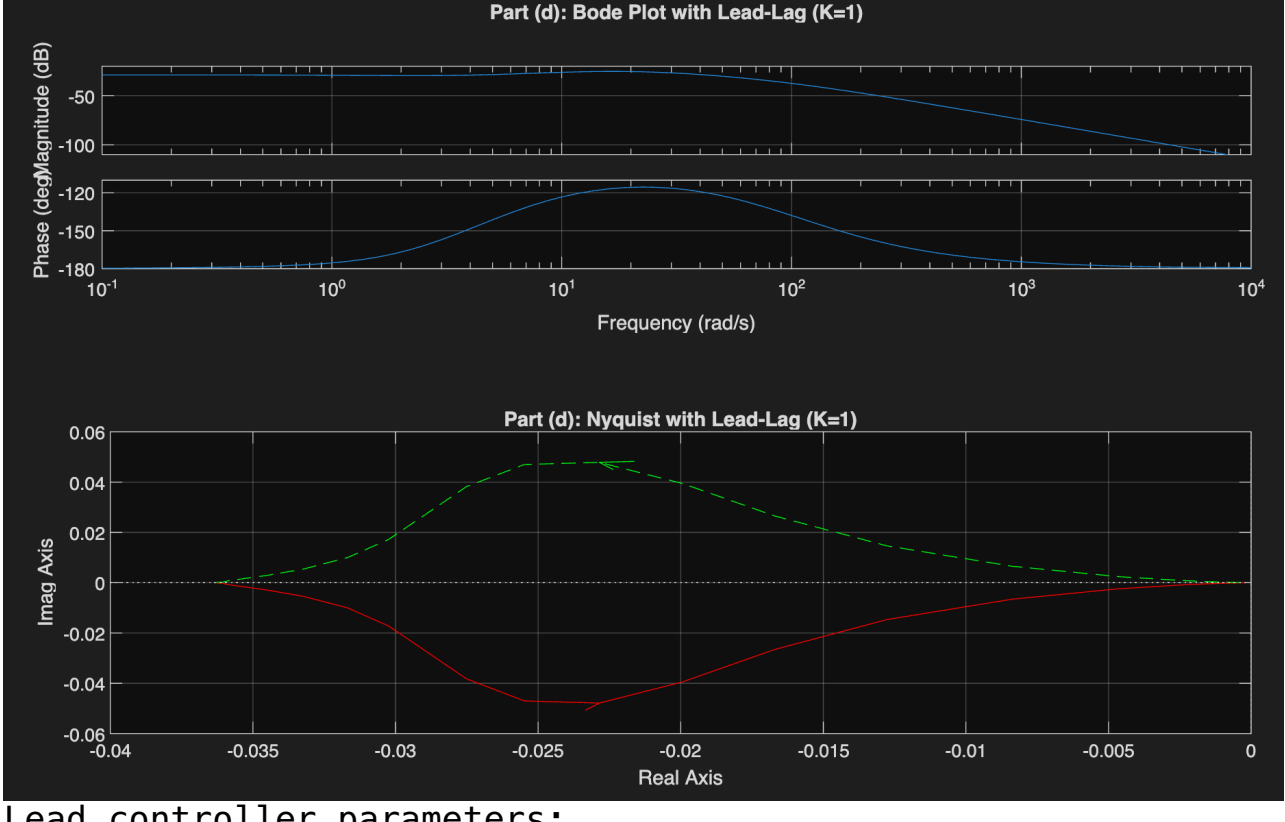
3b. Currently there are none. There should be 1 because there is 1 rhp pole at 18.3769

3c.



It made the nyquist plot smaller, changes the circle shape and shifts the plot to the left. The most important change is that it encircles the real axis where before it had an asymptote there.

3d. Lead lag controller implemented bode and nyquist plots

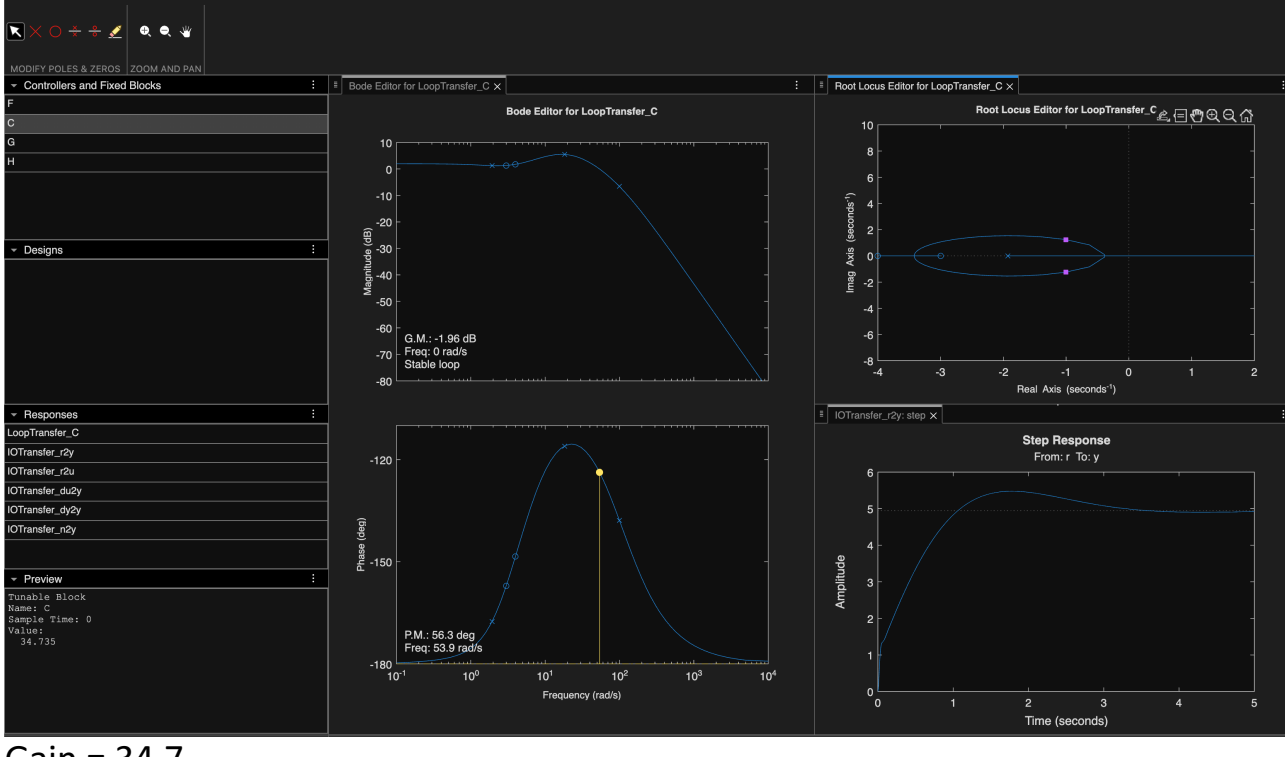


Lead controller parameters:
z_lead = 4.00 rad/s
p_lead = 100.00 rad/s
Maximum phase occurs at w = 20.00 rad/s

Maximum phase lead Phi = 67.38 degrees

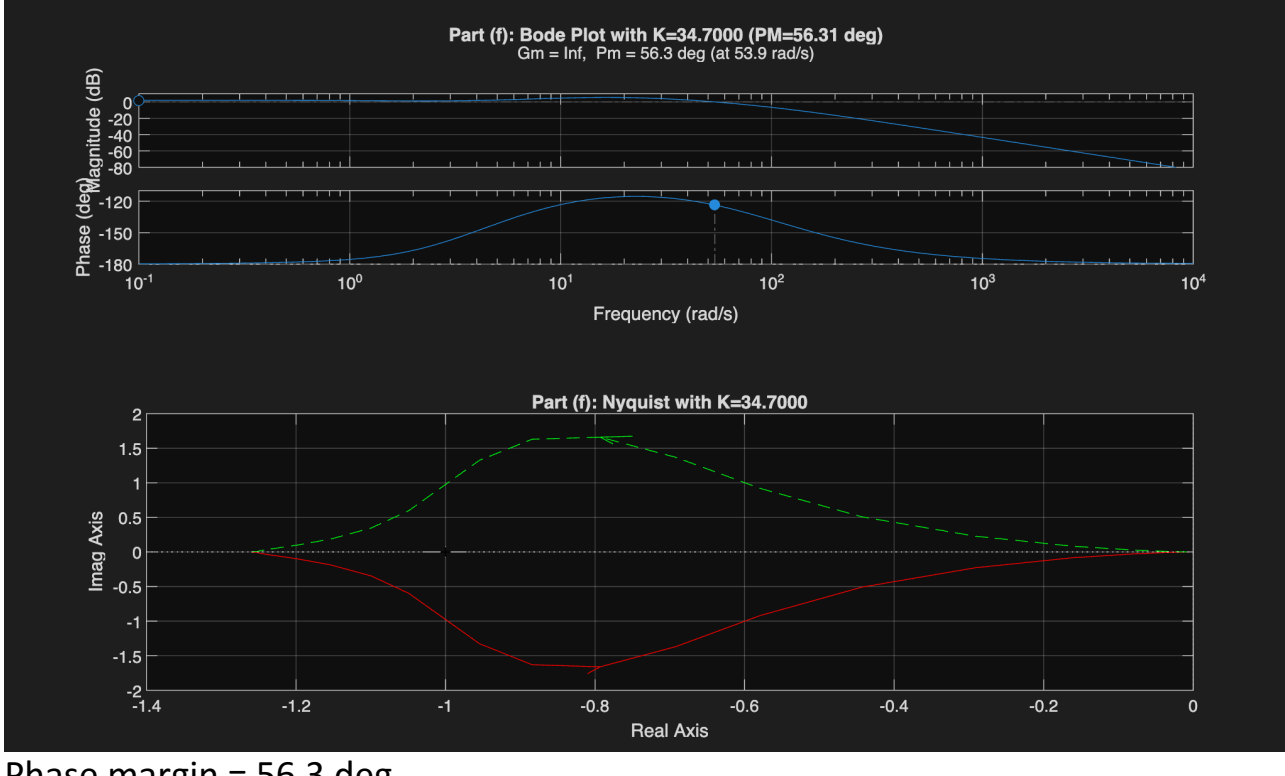
3e. Used control system designer and root locus plot to figure out what gain to use

Moved the poles to the left until there are no rhp poles



Gain = 34.7

3f. Bode and Nyquist plot with Lead Lag and Gain tuned to be stable



Phase margin = 56.3 deg

3g. Impulse response to final design

